



INSPECTION STANDARD (HT LAB)

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☐ Prototype	☐ Pre-Launch	☒ Production	Date (Original)		11/08/22	
Part Name:- SPINDLE-WHEEL		Part No.:- 142-8724		Inspection Standard Number		SIP/3137/250
Customer :- CATERPILLAR		Change Level.:-00		Part Stage		HT Lab.

Sr.No.	Parameter	Special Char. Class	Methods				Reaction Plan
			Product Specification	Inspection Method	Inspection Frequency	Control Method	
1.	Surface Hardness at Location A,B,C,D,E	*	47 HRC min.	HRC / Micro Vicker Hardness Tester	1Pc/Lot	HT Lab Report	A
2.	Surface Hardness at F and G	*	33 HRC min.	HRC / Micro Vicker Hardness Tester	1Pc/Lot	HT Lab Report	A
3.	Soft Zone From Face	*	10 max.	Etching+ HRC / Micro Vicker Hardness Tester	1Pc/Lot	HT Lab Report	A
4.	ECD Below Root 24 mm From Face of Spline At Location A	*	2.5 mm min	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
5.	ECD at (42HRC min.) on Outer Diameter Ø51.10. At Location B	*	6.0~9.0 mm	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
6.	ECD at (42HRC min.) on Outer Diameter Ø49.70 At Location C	*	6.0~9.0 mm	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
7.	ECD at (42HRC min.) on Outer Dia(Ø51.10) At Location D	*	6.0~9.0 mm	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
8.	ECD measured at Fillet Radius On 45° at Location E	*	6.0 mm min.	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
9.	ECD measured at Fillet Radius On 45° at Location F	*	3.0 min.	Micro Vicker Hardness Tester, Vicker Hardness Tester/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
10.	Surface Hardness 33 HRC At Location G, Induction Length From Ø62.0	*	6mm min.	Etching+Vernier Calliper/Rockwell Hardness Tester	1Pc/Lot	HT Lab Report	A
11.	Case Micro structure at Location A		Fully Martensite Structure upto 2.0 mm Depth from Surface at 100X Magnification	Microscope	1Pc/Lot	HT Lab Report	A
12.	Case Micro structure at Location B,C, D and E		Fully Martensite Structure upto 3.0 mm Depth from Surface at 100X Magnification	Microscope	1Pc/Lot	HT Lab Report	A
13.	Core Micro		Uniform Distribution of Perlites and Ferrites	Microscope	1Pc/Lot	HT Lab Report	A
14.	Austenitic Grain Size After Induction		Austenitic Grain Size in Martensitic condition shall be ASTM 5 or finer as per ASTM E112 In Non Uniform Areas where Over Heating is difficult to Avoid, ASTM 3 is acceptable	Microscope	1Pc/Lot	HT Lab Report	A
Note			For Exceptions, Refer to Standard Number: 1E0288A				

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SR NO 01

ISSUED TO U-7

DATE 4/11/2024

Reaction Plan: A-यदि Inspection करते समय कम्पोनेंट खराब मिले तो खराब कम्पोनेंट को Rejection का टैग लगाकर Rejection टेबल पर रखें और उसके बारे में Department के Incharge को बतायें।

Legends: ☒ Special Characteristics as Per Drawing ☒ Special Characteristics For Process ☒ Special Characteristics For Safety

Change Details							
Date	Rev. No.	Change	Reason for Change	Date	Rev. No.	Change	Reason for Change

Prepared By:

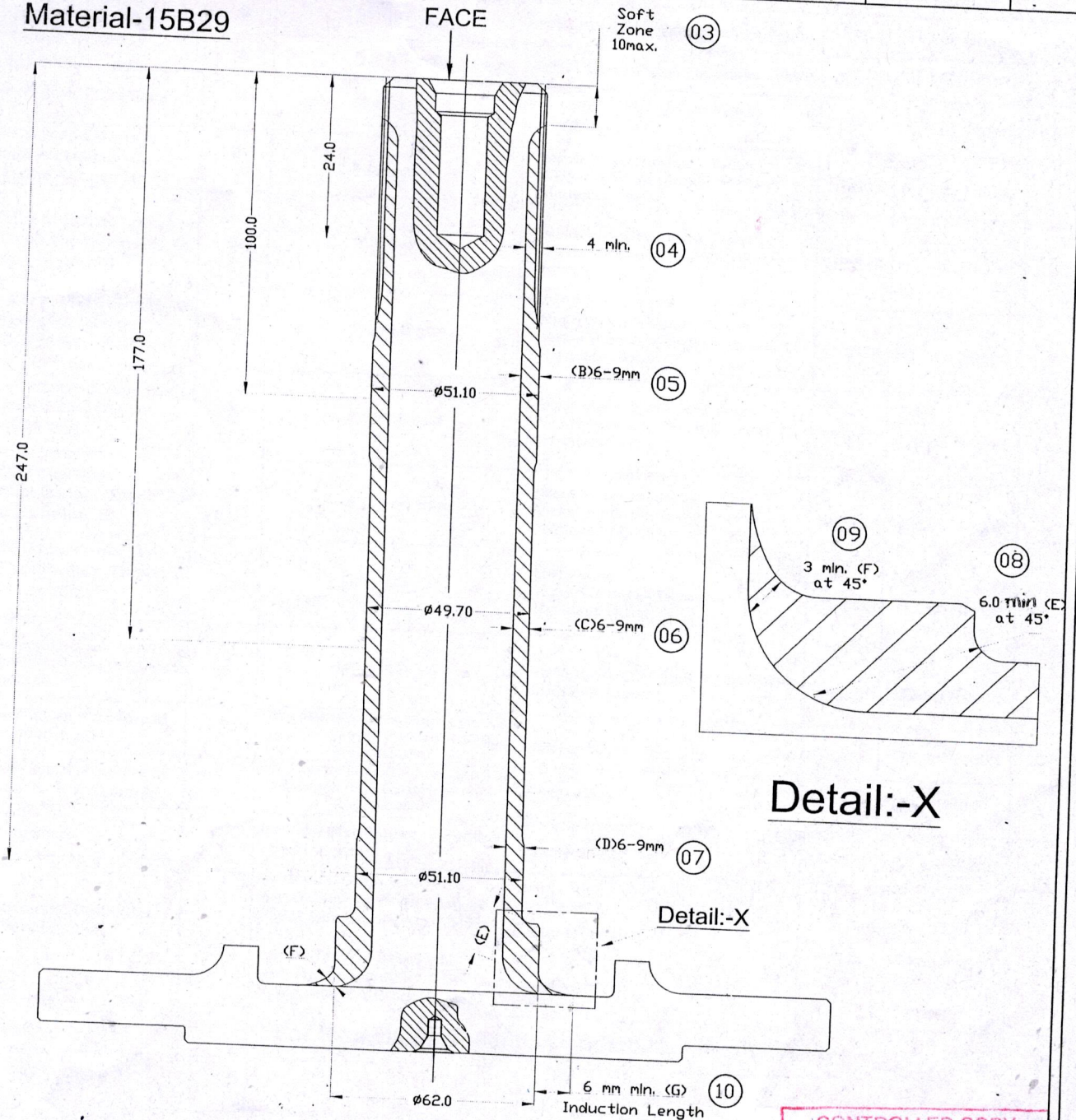


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Material-15B29



Detail:-X

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